Raphael Pelerin



Software Engineering Intern · Application Development, Front-end, Back-end, Full-stack — available from March 2027

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Summary

Fifth-year **Computer Engineering** student at Epitech (Master's, graduating 2027) applying for the **Technical Internship: Information Systems and Technology**. Programming experience in **Python** (server-side: FastAPI, Flask), **JavaScript** and TypeScript, **HTML** and **CSS3** (web development), and C++. I have shipped **full-stack** products, led the web client of a product with real users, completed two software engineering internships (**related internship experience**) and worked in a remote team spread over three countries. I measure before I decide (quantitative habits) and I work directly with the people who use what I build (interpersonal skills). Bilingual in English and French, with a year studying in Beijing. Drawn to quality engineering, data engineering and site reliability as much as to application development.

Technical Skills

Languages	Python, JavaScript, TypeScript, HTML5, CSS3, C++, SQL, Assembly
Web & application development	React, React Native, Angular, Ionic, Expo, Tailwind CSS, Vite, REST APIs, component-driven front-end development
Server-side & back-end	Python (FastAPI, Flask), Node.js, Express, Prisma, REST, SOAP/XML
Data engineering	PostgreSQL, MongoDB, MySQL, SQL
Quality engineering	pytest, automated testing, unit testing, GitLab CI
DevOps & site reliability	Docker, Git, GitLab CI, Unix (Epitech DevOps specialisation)
AI & machine learning	PyTorch, TensorFlow, diffusers/SDXL, ControlNet, OpenAI API, Claude API, Gemini

Experience

Appolonia — Front-end Software Engineer, Intern — Apr – Aug 2025
Casino floor management systems (production)
- Rebuilt the **KT Updater** module as a component-driven admin panel (Angular, TypeScript) automating terminal shutdown and locking — production software where a UI mistake takes a slot machine offline.
- Shipped the same feature set across **Angular (web) and Ionic (hybrid mobile)** from one codebase, consuming SOAP/XML APIs; worked in GitLab.
- Built an internal email-generation tool from scratch: templating, live preview and send flow.

ARD — Mobile Application Development, Intern — 2023
NFC access-control software
- Built an Angular application managing **NFC cards** and user profiles for alarm-system installers, from component structure to state.
- Implemented the **card-scanning flow**: a tag read renders the cardholder's information in the UI in real time.
- Owned the interface end to end, working directly with the installers who used it daily.

Intel AI for Youth — competition — 2022
- Computer vision and object recognition in Python with TensorFlow. **2nd place**, team category.

Projects

Syntheza — Lead Frontend Web (leadership) — 2025 – 2026
syntheza.ovh · remote team of 3 across three countries
- **Led the web client**: interface, design system and React component layer, kept to a ~330 kB gzipped bundle. Stack: React 19, TypeScript, Tailwind, Node.js, Express, Prisma, PostgreSQL.
- Scope set by a field study of 20 professionals before any code was written: **85% reported daily information overload** and 70% lose one to two hours a day sorting noise.

Forge — solo, open-source — 2026
github.com/RaphaelPelerin/forge-asset-generator
- **Python back-end**: FastAPI REST service running SDXL and ControlNet on a single local 8 GB GPU, fully offline, with a React front end. ~4,100 lines.
- **59 automated tests (pytest)** that run in about 5 seconds with no GPU; a test fails any recipe that exceeds the 75-token prompt budget. Measured failure modes (French prompts failed 6/6) before fixing them.

Keys GPT — solo, open-source — 2025 – 2026
github.com/RaphaelPelerin/keys-gpt
- JavaScript/React web app: describe a part in text, get a playable piano roll and a downloadable MIDI file via the OpenAI API.

Keys — VST3 sampler, solo, open-source — 2025
github.com/RaphaelPelerin/keys-gpt-vst
- C++/JUCE instrument with 16 voices, ADSR envelope and filter; every control drawn by hand with the graphics API, no GUI library.

Education

Epitech Technology — Master's, Computer Engineering, Nice (graduating 2027) — 2022 – 2027
Functional and object-oriented programming, Unix, assembly, systems and network programming. Specialisations: DevOps, AI, web/mobile, cybersecurity.

Beijing Jiaotong University — fourth year abroad, software engineering — 2025 – 2026

Lycée Simone Veil — Baccalaureate: Computer Science (NSI) and Mathematics — 2022

Languages

English bilingual (working language) · **French** native · **Mandarin** HSK 3 · **Italian** basic